

Yassine Abassi

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EDUCATION

Polytechnique Montréal

Bachelor's in Computer Engineering

Montreal, QC

Aug 2022 – May 2026

EXPERIENCE

Software QA Automation Intern

May 2024 – Sep 2024

UpToTest

Ariana, Tunisia

- Architected a Cypress/Cucumber BDD test framework using the Page Object Model, implementing **400+** end-to-end scenarios across core user journeys; integrated the suite into **GitLab CI** for automated regression, achieving **89%** feature coverage.
- Executed cross-browser test runs against a dedicated pre-production environment and configured **Mochawesome** HTML reports delivered to stakeholders every sprint, providing full run visibility and enabling data-driven release decisions.
- Collaborated within an Agile Scrum team to align QA priorities with release milestones and progressively shift testing earlier in the development cycle.

Private Tutor

Aug 2021 – Present

Self-employed

Montreal, QC

- Provided in-person tutoring in Mathematics, Physics, and Programming to **20+** college and university students, consistently helping students achieve noticeable grade improvements through tailored explanations and structured problem-solving sessions.

PROJECTS

Hydro-Québec Weather Forecasting Platform

- Built a full-stack weather visualization platform using Angular and ASP.NET Core, featuring multi-model weather display with interactive geospatial mapping via Leaflet and real-time updates via SignalR.
- Implemented GDAL raster processing for cartographic rendering; designed RESTful APIs backed by an SQLite database via EF Core, containerized the application with Docker, and set up a CI/CD pipeline for automated testing and deployment.

3D Interactive Portfolio

[Live Demo](#) 📄

- Crafted a 3D portfolio in React.js and Three.js featuring an interactive room with a functional OS, custom camera rigging, and state-driven transitions across mobile and desktop.
- Expanded the user experience with terminal emulation, procedural art, and embedded mini games. Integrated a Gemini-powered chatbot assistant with Upstash Redis-backed rate limiting.

SSH Terminal Portfolio

[Live Demo](#) 📄

- Built an SSH-based interactive terminal portfolio in Go using the Charm ecosystem, featuring animated splash screens, keyboard navigation, and dynamic theme switching.
- Deployed on Fly.io with Docker containerization, persistent SSH host key management via mounted volumes, and automated CI/CD through GitHub Actions.

Corporate LinkedIn Translator

[Github](#) 📄

- Built and published a Chrome extension that uses Google Gemini AI to detect and translate LinkedIn corporate buzzwords into plain satirical language with inline tooltip annotations.
- Architected a full Node.js/Express backend API deployed on Fly.io with rate limiting, graceful error handling, and dynamic DOM injection to handle LinkedIn's React SPA structure.

WildGuard – AI Wildlife Monitoring System

[Github](#) 📄

- Built an AI-powered ecological surveillance platform in Python for anti-poaching and endangered species protection, with real-time detection updates via Socket.IO and automated SMS alerting through Twilio.
- Implemented a YOLOv8 detection model capable of identifying 50+ species, integrated with OpenCV for video processing and a Flask backend for serving the application.

Multi-Robot Exploration System

- Designed a coordinated multi-robot exploration system for AgileX Limo robots using ROS2 and Gazebo for simulation, implementing custom nodes for inter-robot communication and task distribution.
- Implemented navigation and inter-robot coordination in Python while containerizing software modules using Docker for reproducible deployment across physical and simulated environments.

Embedded Obstacle Detection System

- Programmed a fully autonomous mobile robot in C++ on an ATmega164A microcontroller, implementing low-level hardware control including PWM motor driving, sensor polling, and interrupt-driven decision logic.
- Designed obstacle detection and path-following algorithms using real-time sensor feedback.

TECHNICAL SKILLS

Languages: C, C++, C#, Java, Python, JavaScript, TypeScript, Go, Kotlin, Swift, VHDL, Assembly

Web Frameworks: Angular, React.js, Node.js, NestJS, Express.js, WebSockets, SignalR, .NET, Three.js

Data & APIs: PostgreSQL, MongoDB, SQLite, Firebase, Retrofit, OkHttp, EF Core, GDAL, RxJS, Upstash Redis

DevOps & Tools: Git, GitHub Actions, GitLab CI, Docker, Jira, Fly.io, Qt, Cypress, Vivado

Robots & Boards: Zynq7000, ATmega164A, Pynq-Z1, Basys3 Artix-7, AgileX Limo